SO WE THOUGHT... HOW DO WE CREATE MORE IMPACT?

So we added dataset from youtube comments about similar topics!

Objective 1 — Cross-Register Pattern Distribution

- Construct a comparative frequency table of seven linguistic patterns across the formal and informal corpora, identify which patterns are register-stable versus register-sensitive, and whether informal social media Hindi preserves, suppresses, or transforms each pattern relative to broadcast speech.

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Objective 2 — System Comparison: Regex vs. Neural (Stanza)

- Evaluate rule-based regex extraction against a Stanza-based neural pipeline on both corpora independently, reporting per-pattern precision, recall, and F1. This comparison serves two purposes: first, to establish whether hand-crafted rules remain competitive with a neural system on well-formed formal text; and second, to measure how gracefully each approach degrades when applied to the noisier, code-switched informal register.

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Objective 3 — Identifying Lightweight Register Markers

- Determine whether any subset of the seven patterns functions as a reliable, computationally cheap signal for register classification, that is, whether a pattern's presence or frequency alone can distinguish formal from informal Hindi without requiring a trained classifier. A pattern qualifies as a strong register marker if it shows high distributional divergence between corpora and high regex F1 in at least one register, making it actionable without a neural pipeline. This finding would have practical implications for low-resource or real-time NLP applications on Hindi text.

COMPUTATIONAL LINGUISTICS - 1 PROJECT

vidushi agarwal, ananya pai

Do linguistically motivated regex rules for formal Hindi broadcast speech generalize across registers, and what do the failure modes reveal about register-specific variation?

PROGRESS MADE SINCE MID-SUBMISSION

- Considering social media dataset, finding out the patterns and doing annotation for the complementary dataset.
- Manual annotation of 600-800 sentences randomly chosen from the datasets for all seven linguistic patterns, across both our datasetsc
- Precision, recall, and F1-scores were computed for each pattern using the annotated dataset.
- Conducting error analysis for both the pipelines for both the datasets and doing final comparions

THE OTHER DATASET

- scraped from youtube comments under youtube videos about mann ki baat's so that the topics are similar and hence relevance ensures our results are only affected by the register.
- ran a primitive normalization pipeline involving removal of emojis and languages that are not hindi or english.

```
अभी 1 साल पुराना बात है
बिलकुल ताज़ा खबर के लिएबधाई
ये किसने कह दिया कि फिर शपथ लेंगे?
बहुत खूब अब खेल शुरू होगा समय का
अच्छी बात है जो होता है अच्छे के लिए होता है किसी के लिए कर ना शक तो उसको भोगना पड़ेगा
मैने सोचा की अच्छे दिन आ गए
Ye to खुश खबरी सुना दी
बहुत सुन्दर
good बुरे दिन जायेगे
बहुत खुशी की बात है
खुशी की बात है
सत्यमेव जयते
अब तो बारिस बंद हो जाएगी हवा बंद हो जाएगी अब देश कैसे जियेगा
बहुत सही हुओ
बहुत सुन्दर किया है
भाई साहब तो जब तक जान है तब तक मैं सता पे रहेगा
देश बच गया
बहुत बहुत बहुत खुशी की बात है अब देश का भला होगा
बहुत अच्छा हुआ .
```

MANUAL ANNOTATIONS

- The social media and broadcast speech annotations have been done separately to generate gold data sets corresponding to two different registers.
- Total sentences annotated: ~725
- For each sentence, annotations have been done for all 7 linguistic patterns. These annotations were done separately by both the team members and discussed in order to produce the most accurate metrics.
- Use case of such an annotation:
    - analysis within the same contextual unit
    - enabling accurate co-occurrence analysis
    - reliable metric computation
    - precise error identification.

CALCULATION OF METRICS AND ERROR ANALYSIS

- Used gold standard manually annotated dataset to conduct evaluation.
- Precision, Recall and F1 score have been calculate for stanza based and regex based pipeline using a basic python script for both Social media text and Broadcast speech

Purpose:

- Improves precision/recall/F1 computation by keeping gold and predicted units consistent.
- Makes error analysis more interpretable, as errors can be traced to specific sentences.
- Helps compare register-based variation (social media vs broadcast speech) at a meaningful unit of analysis

RESULTS

```
T1_076
राजस्थान में सीकर के रामदेव सिंह जी को kidney transplant कराना पड़ा था .
Reduplication: 0 | Conjunctive Verb: 1 (कराना) | Compound Verb: 1 (कराना+पड़ा) | Honorific: 0 | Acronym: 0 | MWE: 0 | Hyphen Pair: 0

T1_077
आज वे sporting activity में कमाल कर रहे हैं .
Reduplication: 0 | Conjunctive Verb: 1 (कर) | Compound Verb: 0 | Honorific: 0 | Acronym: 0 | MWE: 0 | Hyphen Pair: 0

T1_078
साथियो, आपको इस तरह के बहुत से प्रेरक उदाहरण देखने को मिल जाएंगे.
Reduplication: 0 | Conjunctive Verb: 0 | Compound Verb: 2 (देखने+जाएंगे, मिल+जाएंगे) | Honorific: 1 (आपको [formal]) | Acronym: 0 | MWE: 0 | Hyphen Pair: 0
```

```
T1_014
दुनिया भर के leaders ये देखकर हैरत में पड़ गए कि कैसे AI की मदद से हम हमारे प्राचीन ग्रंथों को, हमारे प्राचीन ज्ञान को, हमारी पांडुलिपियों को संरक्षित कर रहे हैं, आज की generat
Reduplication: 0 | Conjunctive Verb: 1 (dēkha-kar) | Compound Verb: 1 (dēkhakar+paṛ) | Honorific: 0 | Acronym: 1 (AI) | MWE: 0 | H

T1_015
साथियो, Exhibition के दौरान display के लिये सुश्रुत संहिता का चयन किया गया।
Reduplication: 0 | Conjunctive Verb: 0 | Compound Verb: 0 | Honorific: 0 | Acronym: 0 | MWE: 1 (kē liyē) | Hyphen Pair: 0

T1_016
पहले step में दिखाया गया कि कैसे technology की मदद से हम पांडुलिपियों की image quality सुधारकर उन्हें पढ़ने लायक बना रहे हैं।
Reduplication: 0 | Conjunctive Verb: 1 (sudhāra-kar) | Compound Verb: 1 (step+gayā) | Honorific: 0 | Acronym: 0 | MWE: 0 | Hyphen
```

REGEX VS STANZA AGREEMENT - FORMAL

Pattern	Regex	Stanza	Stanza/Regex ratio	Agreement
Reduplication	36	65	1.81×	Moderate
Conjunctive Verbs	91	103	1.13×	High
Compound Verbs	93	118	1.27×	High
Honorifics	25	72	2.88×	Low
Acronyms	33	33	1.00×	Perfect
MWEs	72	124	1.72×	Moderate
Hyphenated Pairs	21	67	3.19×	Low
Total	**371**	**582**	**1.57×**	

REGEX VS STANZA AGREEMENT - INFORMAL

Pattern	Regex	Stanza	Stanza/Regex ratio	Agreement
Reduplication	4	38	9.50×	Very Low
Conjunctive Verbs	34	65	1.91×	Moderate
Compound Verbs	53	81	1.53×	Moderate
Honorifics	15	21	1.40×	Moderate
Acronyms	6	6	1.00×	Perfect
MWEs	21	20	0.95×	Near-Perfect
Hyphenated Pairs	2	3	1.50×	Moderate
Total	**135**	**234**	**1.73×**	

WHAT DOES OUR DATA TELL US?

Pattern	Formal (raw)	Formal /1k	Informal (raw)	Informal /1k	Ratio F:I
Reduplication	65	6.66	38	8.98	0.74×
Conjunctive Verbs	103	10.55	65	15.37	0.69×
Compound Verbs	118	12.09	81	19.15	0.63×
Honorifics	72	7.37	21	4.96	1.49×
Acronyms	33	3.38	6	1.42	2.38×
MWEs	124	12.70	20	4.73	2.69×
Hyphenated Pairs	67	6.86	3	0.71	9.67×
Total	**582**	**59.63**	**234**	**55.32**	**1.08×**

OVERALL RESULTS

Rank	System + Corpus	Overall F1
1	Neural — Comments	0.8645
2	Neural — Broadcast	0.8645
3	Regex — Comments	0.5929
4	Regex — Broadcast	0.3973

- Regex is surprisingly competitive on formal text - Across the formal corpus, regex captures 64% of what Stanza finds (371 vs 582 total instances). For structured patterns like acronyms (perfect agreement) and conjunctive verbs (1.13× gap), regex is essentially equivalent to a neural pipeline at a fraction of the computational cost. This validates the original intuition that Hindi's predictable morphology is regex-friendly, but only when the input is clean and well-formed.

COMPOUND VERBS

System	Corpus	Compound Verb F1	TP	FP	FN
Neural	Broadcast	0.4122	81	14	217
Regex	Broadcast	0.3680	62	10	203
Neural	Comments	0.9011	41	0	9
Regex	Comments	0.6353	27	8	23

- This is the most counterintuitive result in the entire study. Both systems perform worse on compound verbs in formal broadcast speech than in informal comments, neural drops from 0.90 to 0.41, regex from 0.64 to 0.37. The FN counts (217 and 203) are catastrophic relative to TP counts. The likely cause is that broadcast Hindi uses more complex, embedded compound verb constructions where the vector verb is separated from the stem by intervening material, which neither system handles.

REDUPLICATION

System	Corpus	F1	Precision	Recall
Neural	Broadcast	0.6392	0.544	0.775
Regex	Broadcast	0.2857	1.000	0.167
Neural	Comments	0.9474	0.900	1.000
Regex	Comments	0.9032	0.966	0.848

- Regex reduplication on broadcast is precision-perfect but recall-devastated (0.167), it finds only 3 of 18 instances but never fires incorrectly. On comments, both systems perform excellently (0.90+ F1). This inverts the expected pattern, you'd expect formal text to be easier for regex.

CONJUNCTIVE VERBS

System	Corpus	F1	Precision	Recall	FP
Neural	Broadcast	0.4490	0.328	0.710	45
Regex	Broadcast	0.6316	0.581	0.692	13
Neural	Comments	0.4952	0.366	0.765	45
Regex	Comments	0.7097	0.611	0.846	14

- Regex outperforms Stanza on conjunctive verbs in both registers. The key difference is precision, regex generates 13 FPs on broadcast versus Stanza's 45, and 14 FPs on comments versus Stanza's 45. Stanza achieves marginally higher recall in both corpora but at the cost of more than three times the false positives. For downstream tasks where false positives are costly, regex is the recommended system for this pattern in all conditions.

ACRONYMS

System	Corpus	F1	Precision	Recall
Neural	Broadcast	0.8889	1.000	0.800
Regex	Broadcast	0.9057	1.000	0.828
Neural	Comments	0.7692	1.000	0.625
Regex	Comments	0.7692	1.000	0.625

- Precision is perfect (1.000) across all four conditions. Regex outperforms Stanza on broadcast acronyms (0.91 vs 0.89 F1).

MWE'S

System	Corpus	F1	Precision	Recall	FP
Neural	Broadcast	0.2222	0.202	0.247	71
Regex	Broadcast	0.1417	0.184	0.115	40
Neural	Comments	1.0000	1.000	1.000	0
Regex	Comments	0.5294	0.450	0.643	11

- Neural achieves perfect MWE detection on comments (1.00 F1) but nearly fails on broadcast (0.22 F1, 71 FPs). The 71 false positives for neural on broadcast indicate Stanza is over-generating MWE candidates from the dense postpositional structure of formal Hindi.

HONORIFICS

System	Corpus	F1	Precision	Recall	FN
Neural	Broadcast	0.5507	0.864	0.404	56
Regex	Broadcast	0.2667	0.727	0.163	82
Neural	Comments	0.9697	1.000	0.941	1
Regex	Comments	0.6667	0.632	0.706	5

- The 56 FN for neural and 82 FN for regex on broadcast honorifics, against only 38 TP for neural, means both systems are missing more than half the instances. Given that broadcast Hindi is almost exclusively āp-register and the patterns are morphologically regular, this scale of miss suggests either the gold standard annotates a broader category than the system targets, or inflected forms (āpakō, āpanē, āpakī) are not being matched consistently.

CHALLENGES FACED AND HOW THEY WERE ADDRESSED

#1

Challenge: No easily available corpus for informal hindi. corpuses like HASOC and Dakshina dataset were tried but they did not have enough informal data.
Mitigation: scraped youtube comments and ran a normalization pipeline

#2

Challenge: Several patterns have theoretically clear definitions but practically ambiguous boundaries. eg: conjunctive verbs vs postpositions (token + ke), compound verbs vs simple verb sequences, and MWEs vs productive postpositional phrases (when is ke sāth a fixed expression vs a compositional construction).
Mitigation: Discussed and reached a fixed conclusion as to what the boundaries for each of the ambiguity is.

#3

Challenge: Pre-processing, Normalization and Unicode conversions:
Schwa deletion is word-final only; medial schwas remain undeleted (e.g. sarakār instead of sarkār). This would lead to complex rule writing in code.
Mitigation: Instead, it has been ensured that the above rule would be uniformly maintained throughout

#4

Challenge: Calculation of metrics and Error Analysis:
The model was originally very rigid, giving very little accuracy than its actual accuracy.
Mitigation: An attempt to make it less rigid has been carried out by relaxation of exact one-to-one mapping

CONCLUSION

- The key finding is that MWEs and hyphenated pairs are strong register markers, showing formal-to-informal frequency ratios of 2.69× and 9.67× respectively, while compound and conjunctive verbs remain register-stable across both broadcast speech and social media commentary.
- The most linguistically significant contrast appears in reduplication: formal broadcast Hindi uses it as a grammatical distributive and adverbial device (alag-alag, dhīrē-dhīrē, kōnē-kōnē), whereas informal social media Hindi repurposes the same form for rhetorical emphasis and political chanting (mōdī-mōdī, har-har, āzādī-āzādī), highlighting a shift in pragmatic function that frequency alone would not capture.

CONCLUSION

- In system comparison, neural (Stanza) performs better overall, but the gap is smaller than expected: regex achieves 0.40 F1 on broadcast and 0.59 on comments, compared to Stanza's 0.49 and 0.59.
- Crucially, regex counts remain reliably proportional to neural outputs across patterns and registers, allowing meaningful register analysis without a neural pipeline. Regex also outperforms Stanza on conjunctive verbs in both registers (0.63 vs 0.45 on broadcast, 0.71 vs 0.50 on comments), indicating that for morphologically regular, surface-predictable patterns, rule-based methods can remain competitive even on noisy, real-world text.

ERROR ANALYSIS

- overgeneralization of conjunctive verbs - regex
- overgeneralization of reduplication - stanza
- phrases containing honorifics - regex
- compositional MWE's - both
- failure on informal hyphenated pairs - both

OVERALL CONTRIBUTIONS MADE BY US :)

Ananya:

- Unicode conversions and transliterations
- Stanza based pattern identification
- Data Annotation for social media text and Calculation of performance metrics between gold data set and outcomes from both pipelines

OVERALL CONTRIBUTIONS MADE BY US :)

Vidushi:

- Deciding both the datasets
- Regex pipeline
- Discussion on data annotation for social media
- Normalization and obtaining of informal dataset

OVERALL CONTRIBUTIONS MADE BY US :)

Together as a Team:

- The team members individually annotated the sentences from the datasets and after a few discussions and meets, the best out of the two for each of the sentences were chosen.
- Multiple opinions and suggestions were exchanged assessing individual contributions and necessary changes were made by the Team Members on time.

THANK YOU